



**UNIVERSITÀ
DI TORINO**

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Fisica dei Sistemi Complessi

Improving VAE Representational Capabilities for Understanding LLMs

Tesi di Laurea Magistrale

Relatore:

Prof. Michele Caselle

Corelatori:

Prof. Alan Perotti

Prof. André Panisson

Controrelatore:

Prof. Matteo Osella

Candidato:

Francesco Marchisotti

Tutor Aziendale:

Gabriele Ciravegna

Anno Accademico 2025/2026

Francesco Marchisotti: *Improving VAE Representational Capabilities for Understanding LLMs*, Fisica dei Sistemi Complessi

CENTAI TEAM:

Gabriele Ciravegna

Alan Perotti

André Panisson

Francesco Cozzi

UNITO PROFESSORS:

Michele Caselle

Matteo Osella

LOCATION:

Torino

TIME FRAME:

Oct. 2025 - Jul. 2026

“I wish there was a way to know you’re in the good old days
before you’ve actually left them.”

— *Andy Bernard*

ABSTRACT

Short summary of the contents in English...a great guide by Kent Beck how to write good abstracts can be found here:

<https://plg.uwaterloo.ca/~migod/research/beck00PSLA.html>

ZUSAMMENFASSUNG

Kurze Zusammenfassung des Inhaltes in deutscher Sprache...

ACKNOWLEDGMENTS

To whom shall be acknowledged,
you are being acknowledged with this exact statement.

“I’m so funny.”

— The candidate, who’s not actually funny

CONTENTS

1	INTRODUCTION	1
1.1	Motivation	1
1.2	Thesis Structure	1
2	BACKGROUND AND RELATED WORK	3
2.1	NLP	3
2.2	XAI	3
2.2.1	Traditional XAI	3
2.2.2	C-XAI	3
2.2.2.1	Explainable-by-design Models	3
2.2.2.1.1	CBM/CEM	3
2.2.2.1.2	LF-CBM	3
2.2.2.2	Post-hoc Methods	3
2.2.2.2.1	T-CAV	3
2.3	AutoEncoders	3
2.3.0.0.1	Anomaly detection	3
2.3.0.0.2	Denoising AE	3
2.3.1	SAEs	3
2.3.1.1	Architectures	3
2.3.1.1.1	Vanilla ℓ_1	3
2.3.1.1.2	Gated SAE	3
2.3.1.1.3	TopK SAE	3
2.3.1.1.4	JumpReLU SAE	3
2.3.1.2	Usage	3
2.3.1.2.1	MechInt	3
2.3.2	VAEs	4
2.3.2.1	Architectures	4
2.3.2.1.1	β -VAE	4
2.3.2.1.2	VQ-VAE	4
2.3.2.2	Usage	4
2.3.2.2.1	Generative model	4
2.3.2.2.2	XAI	4
3	METHODS	5
3.1	Definitions/Notation	5
3.2	Aim of the Work	5
3.3	Model Development	5
3.3.1	Probabilistic Formulation	5
3.3.2	Architectures	5
4	EXPERIMENTS	7
4.1	Model Training	7
4.2	Evaluation Metrics	7
4.3	Results	7
5	CONCLUSIONS	9

5.1	Contributions	9
5.2	Future Work	9

LIST OF FIGURES

LIST OF TABLES

LISTINGS

ACRONYMS

INTRODUCTION

1.1 MOTIVATION

1.2 THESIS STRUCTURE

BACKGROUND AND RELATED WORK

2.1 NLP

2.2 XAI

2.2.1 *Traditional XAI*

2.2.2 *C-XAI*

2.2.2.1 *Explainable-by-design Models*

2.2.2.1.1 CBM / CEM

2.2.2.1.2 LF-CBM

2.2.2.2 *Post-hoc Methods*

2.2.2.2.1 T-CAV

2.3 AUTOENCODERS

2.3.0.0.1 ANOMALY DETECTION

2.3.0.0.2 DENOISING AE

2.3.1 SAEs

2.3.1.1 *Architectures*

2.3.1.1.1 VANILLA ℓ_1

2.3.1.1.2 GATED SAE

2.3.1.1.3 TOPK SAE

2.3.1.1.4 JUMPRELU SAE

2.3.1.2 *Usage*

2.3.1.2.1 MECHINT

2.3.2 VAEs

2.3.2.1 Architectures

2.3.2.1.1 β -VAE

2.3.2.1.2 VQ-VAE

2.3.2.2 Usage

2.3.2.2.1 GENERATIVE MODEL

2.3.2.2.2 XAI

CONCLUSIONS

METHODS

3.1 DEFINITIONS/NOTATION

3.2 AIM OF THE WORK

3.3 MODEL DEVELOPMENT

3.3.1 *Probabilistic Formulation*

3.3.2 *Architectures*

CONCLUSIONS

EXPERIMENTS

4.1 MODEL TRAINING

4.2 EVALUATION METRICS

4.3 RESULTS

CONCLUSIONS

CONCLUSIONS

5.1 CONTRIBUTIONS

5.2 FUTURE WORK

COLOPHON

This document was typeset using the typographical look-and-feel `classicthesis` developed by André Miede and Ivo Pletikosić. The style was inspired by Robert Bringhurst’s seminal book on typography “*The Elements of Typographic Style*”. `classicthesis` is available for both L^AT_EX and L^YX:

<https://bitbucket.org/amiede/classicthesis/>

Happy users of `classicthesis` usually send a real postcard to the author, a collection of postcards received so far is featured here:

<http://postcards.miede.de/>

Thank you very much for your feedback and contribution.